Problem 8.25. An undirected graph is ***bipartite*** if its nodes may be divided into two sets so that all edges go from a node in one set to a node in the other set. Show that a graph is bipartite if and only if it doesn't contain a cycle that has an odd number of nodes. Let $BIPARTITE = \{\langle G \rangle \mid G \text{ is a bipartite graph}\}$. Show that $BIPARTITE \in \text{NL}$.

Part a. Show that a graph is bipartite if and only if it doesn't contain a cycle that has an odd number of nodes.

Proof. □

Part b. Show that $BIPARTITE \in \text{NL}$.

Proof. □